UKAS QUALITY MANAGEMENT 005

SYSTEM CERTIFICATION ISO 9001:2008 SGS

ACROW

Formwork Technology

Acrow H20 & Soldier System

Acrow MISR Provides Full Site Service Based On More Than 35 Years Experience In The Field

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Introduction

Being one of the largest formwork and scaffolding companies in Egypt, Acrow H20 and soldier system is one of the most flexible systems in Acrow formwork. In addition, it is a multi-purpose system, where it can be used in different applications.

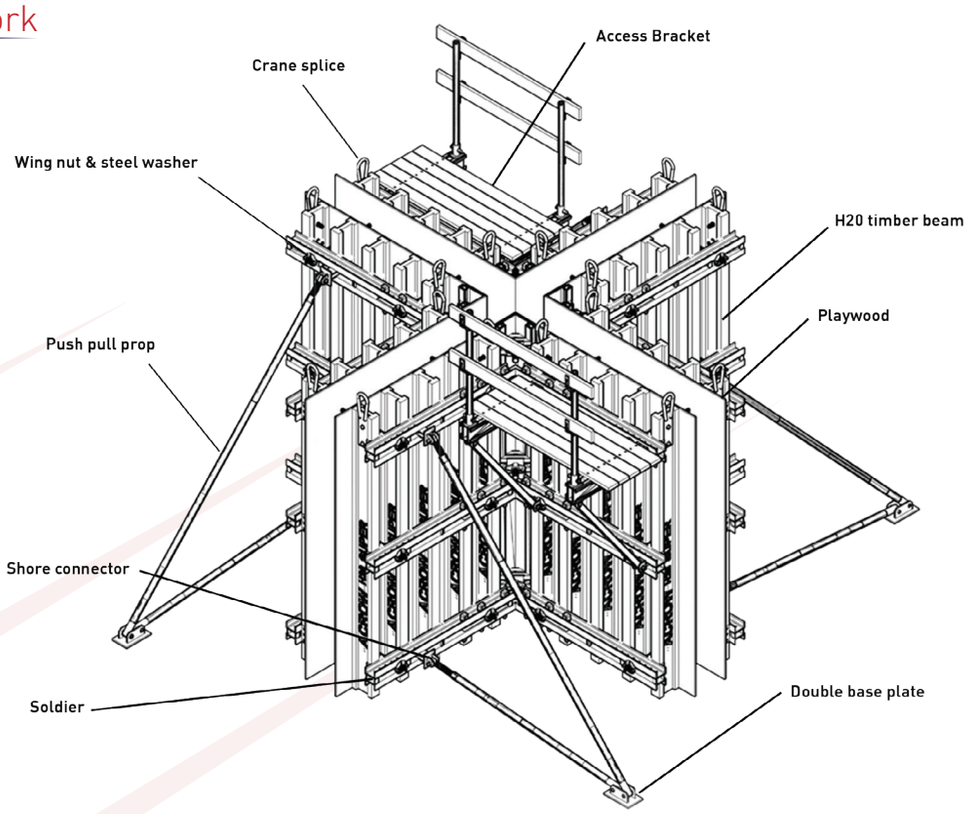
The H20 and soldier wall system formwork elements are easily and quickly assembled by connecting the H20 timber beams to the soldier by the means of H20 connection. Moreover, elements dismantling are done as easily as system erection.

The main H20 and Soldier system advantage is that it provides both high adaptation and easy re-assembling when structure ground plans are changed frequently. In addition, this system proved to be one of the most economical alternatives compared with steel frame formwork panel system, where it comes to both complicated designs and numerous non-typical applications with the same wall heights.

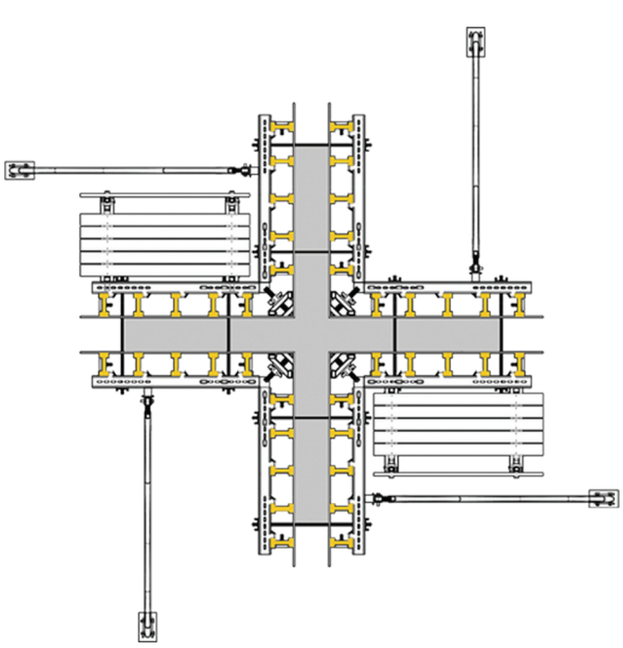
The H20 and Soldier system is typically used for all types of walls and columns where it showed to have high rigidity as well as high stability compared with the panels' low weight.

Product Overview

Wall Formwork



Plan View



Advantages

•Acrow H20 and soldier system is considered to be the most cost effective system that combines between both the timber beams steel soldier and the plywood.

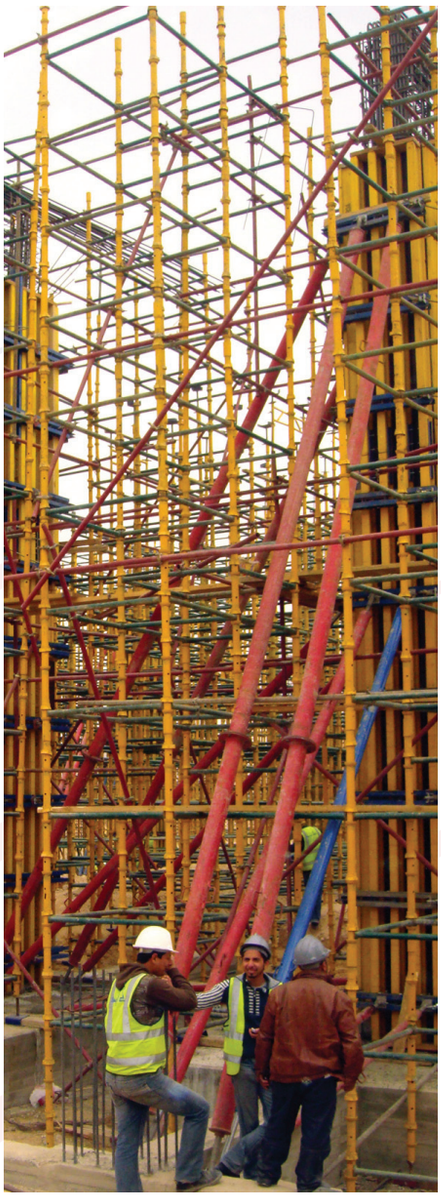
• It gives an architecturally good shape with extremely high quality concrete finishing for the straight and curved walls.

• The high flexural stiffness for timber beams and steel walers speed up shuttering by minimizing the number of used tie rods. It is considered to be an economical system due to the low materials cost.

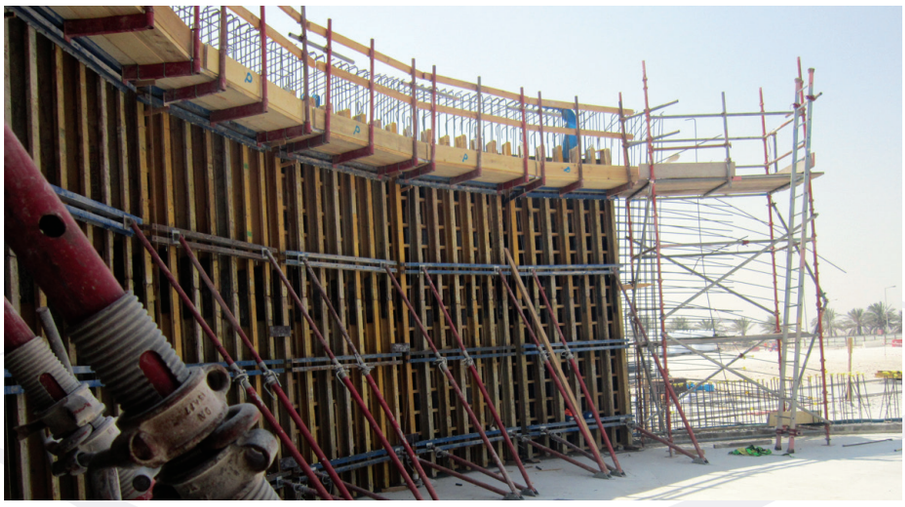
•It is flexible for any height or layout and can be easily shuttered.

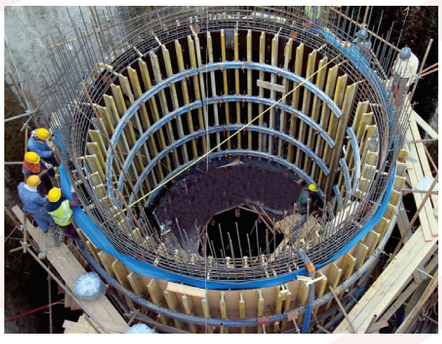
•It is adjustable due to the holes, soldier and splices.

Applications



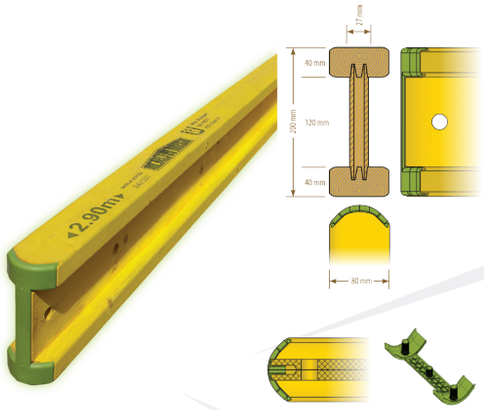
Applications





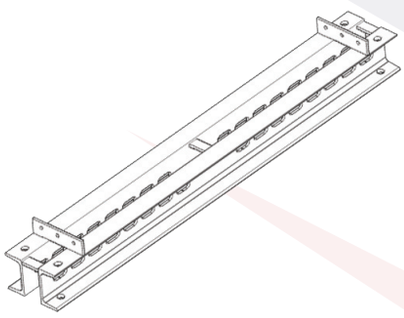
Applications

System Components



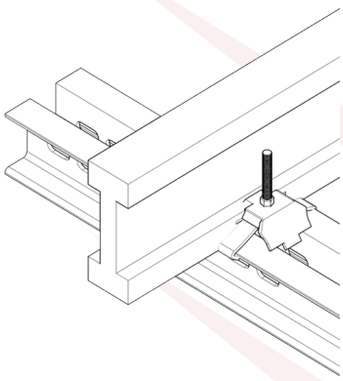
Acrow wooden beams H20

- The new Acrow wooden beam H20 with plastic cap end protection Allowable Values
  Shear force Q = 11 kN
  A support reaction A = 22 kN
  Bending moment M = 5 kNm



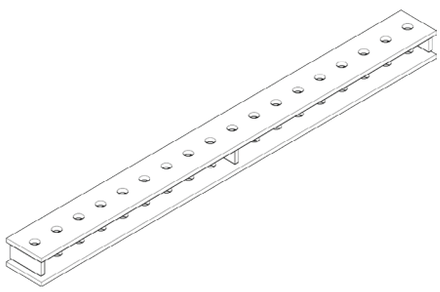
Steel waler (soldier)

- Acrow steel waler provides high accuracy for the panel alignment. Moreover, the slotted end eases the connection to the next panel using Acrow splice.



H20 connection

- H20 connection connects the H20 timber beam with the waler at any required or planned position.



Steel waler splice

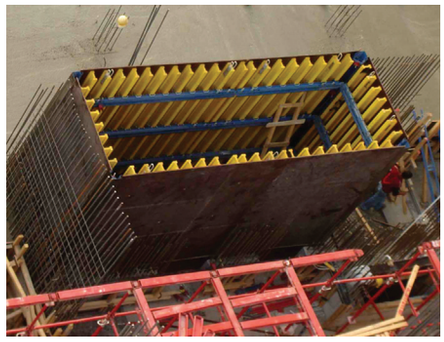
- Using Acrow splice, the two panels can be both easily and fastly connected together. In addition, they will be connected using the same waler splice with Acrow big wedge or where the need for wrench and the fixation time are eliminated.

System Accessories

tilt up shore / push pull prop:

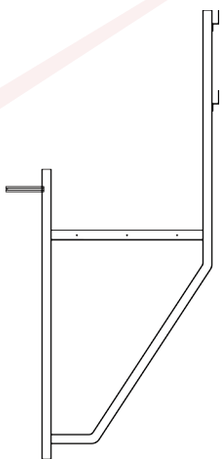
- Tilt up shore is used to adjust both the verticality and the panel location.
- The lower tilt up shore eases the push/pull of the formwork to its right connection.
- The upper tilt up shore eases the panel verticality adjustment.
- The push-pull prop is available for any required length.
- Tilt up shore is available for 5 standard sizes:

- Tilt up shore X0
  Tilt up shore X1
  Tilt up shore X2
  Tilt up shore X3
  Tilt up shore X4



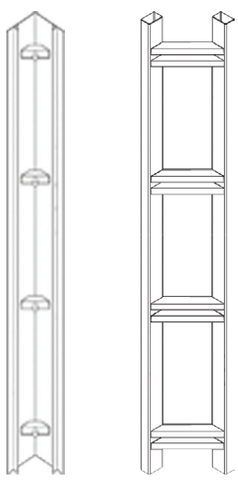
Access bracket

- For safety considerations while concreting, Acrow supplied an access bracket with a handrail post with two levels of timber handrails. The bracket must be connected with two small wedges.



Corner filler panel

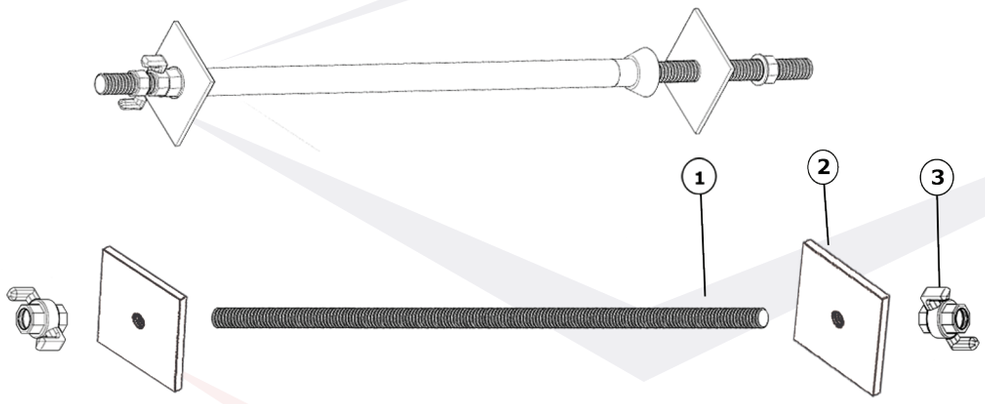
- Acrow corner filler panel eases the shaft form stripping.
- The corner angle is 100% achieved and the stripping is easy.
- Finally, a closed shaft can be done without any filler if corner filler panel was used.



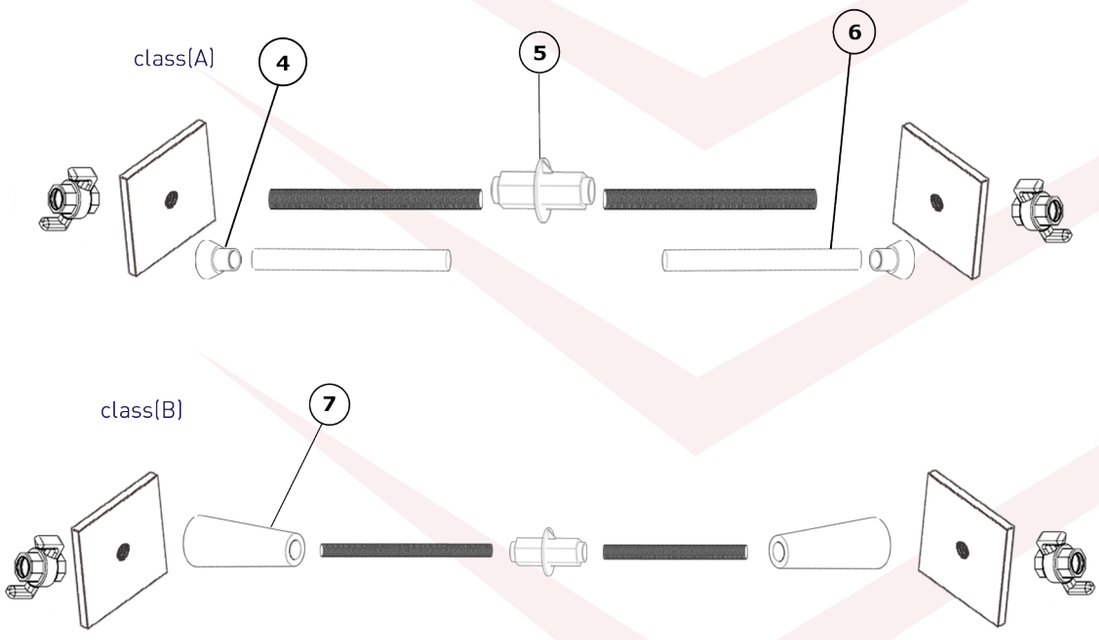
Dw 16 Tie System

The Acrow DW 16-Tie system is primarily used with soldiers requiring minimum number of ties. Moreover, the basis for the assembly is the continuous thread high-tensile steel rod. Finally, the wing nut is designed to freely spin on the fast threaded rod in order to effectively speed the time for the formwork erection and stripping.

1- THRU D W 16 TIE SYSTEM : there is no lost items



2 - WATER SEAL D W 16 tie system



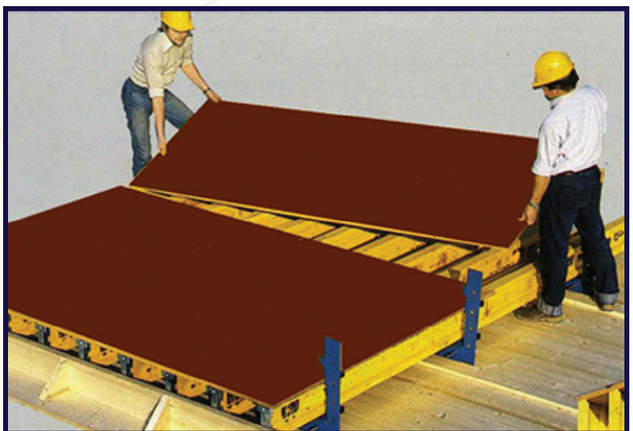
1- D W16 tie rod 2- Steel washer 3- Wing nut 4- Plastic cone 5-Water barrier 6- Palstic tube 7- Steel cone

Formwork Panel Assembly

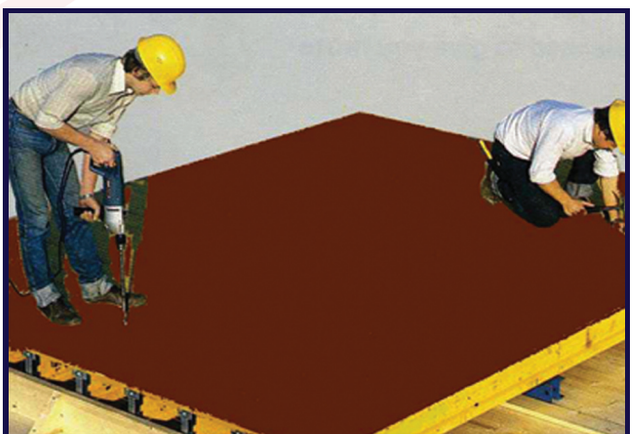
Fix the location of the walers and timber girder spacing on the flat and leveled pattern.

Lay the H20 beam in its' required position.

Connect the timber girder to the steel walers by H20 connection.



Fix the plywood with the timber girder by torx screw.

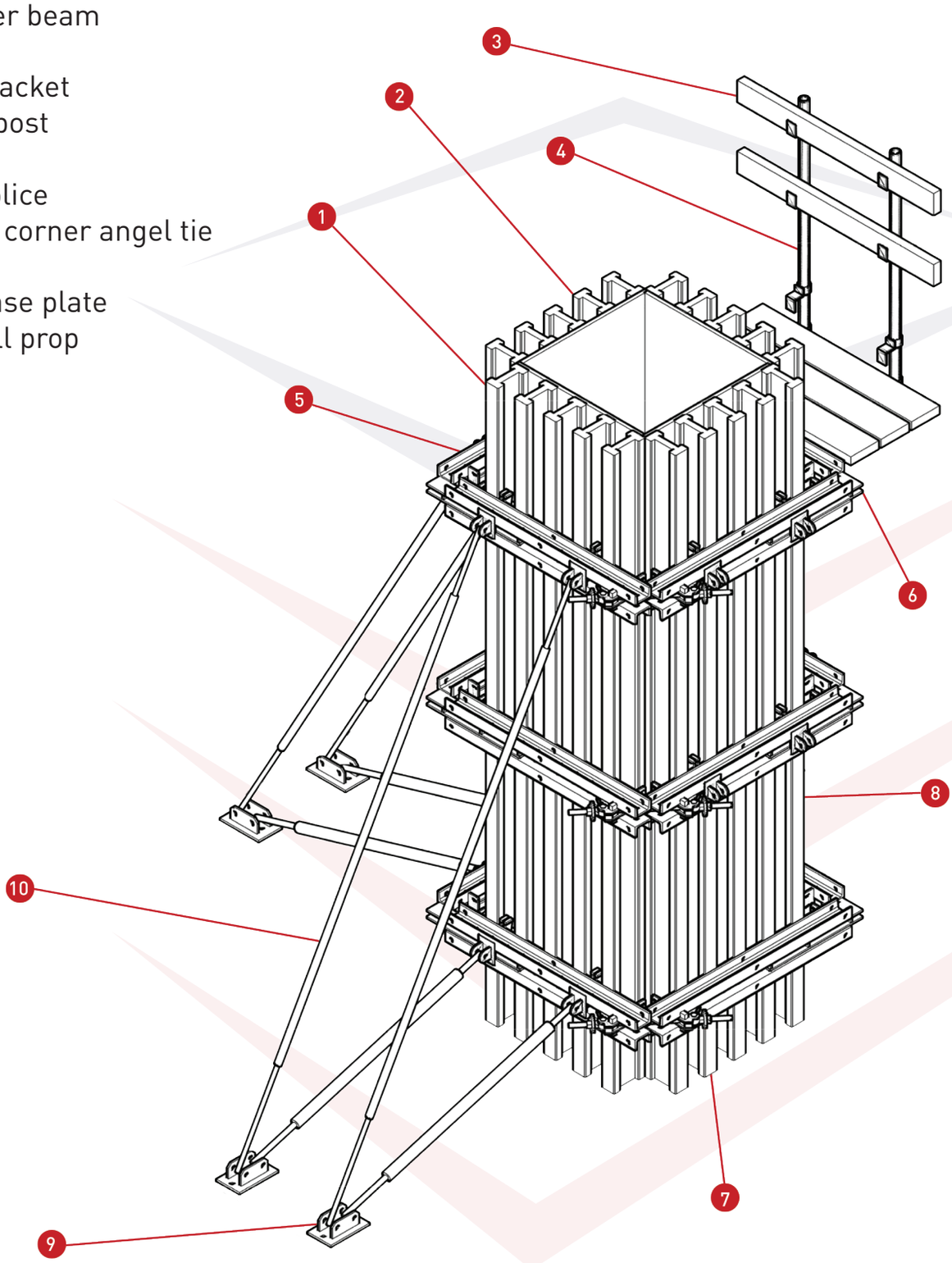


Drill holes for the ties.

Place the panel in the intended position.

Product Overview For Column

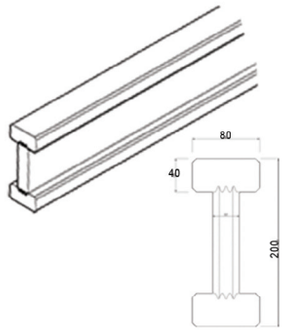
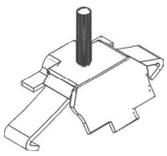
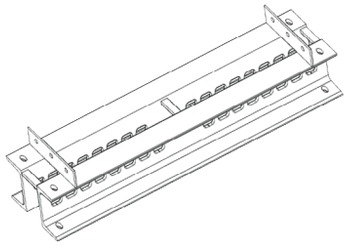
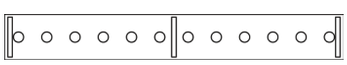
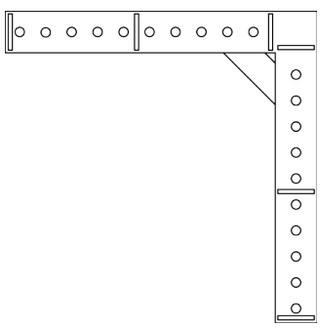
1- H20 timber beam
2- Playwood
3- Access bracket
4- Handrail post
5- Soldier
6- Corner splice
7- Universal corner angel tie
8- Tie rod
9- Double base plate
10- Push pull prop



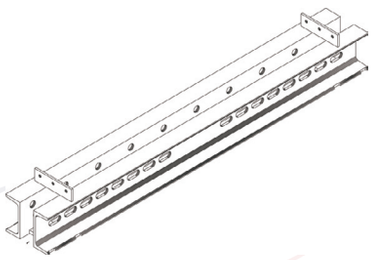
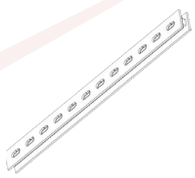
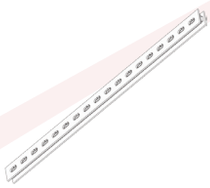
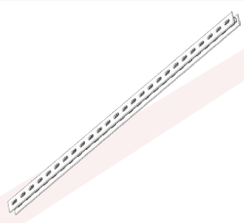
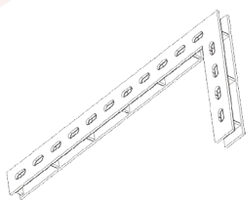
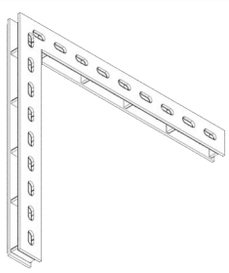
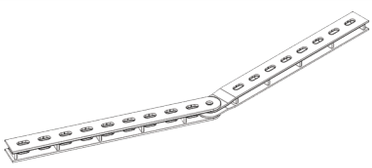
H20 & Soldier System For Columns

- By using the same components of the H20 and soldier wall formwork system in addition to steel column waling, a column formwork can be easily assembled.
- Column formwork is assembled in two L-shaped halves. These two halves are anchored to each other using corner tie connector.
- High permissible fresh concrete pressure makes it easy to pour any column height in one time.
- H20 and soldier system for columns formwork are suitable for both rectangular and circular columns. However, circular columns and other complicated shapes could be achieved by using timber negatives.

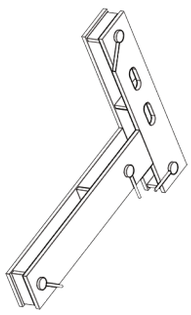
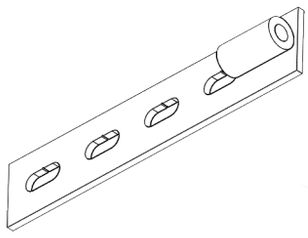
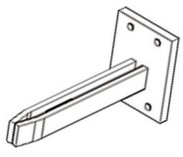
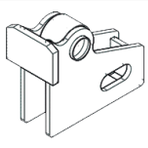
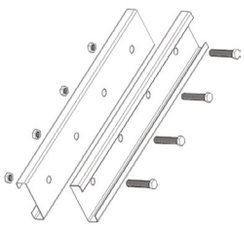
Components

Item	Description	Code no.	Weight (kg)
 Permissible bending Moment = 5.0 KNm Permissible shear for = 11.0 KN Approx. weight = 5.0 Kg/m	Timber H20 L=2.50 m Timber H20 L=2.90 m Timber H20 L=3.30 m Timber H20 L=3.60 m Timber H20 L=3.90 m Timber H20 L=4.50 m Timber H20 L=4.90 m Timber H20 L=5.90 m	6221177024062 6221177024076 6221177024093 6221177024109 6221177024116 6221177024130 6221177024147 6221177024154	12.50 14.50 16.50 18.00 19.50 22.50 24.50 29.50
	H20 connection	6221177006891	0.70
	Soldier][10 75 cm (Holes) Soldier][10 100 cm (Holes) Soldier][10 125 cm (Holes) Soldier][10 150 cm (Holes) Soldier][10 175 cm (Holes) Soldier][10 200 cm (Holes) Soldier][10 225 cm (Holes) Soldier][10 250 cm (Holes) Soldier][10 275 cm (Holes) Soldier][10 300 cm (Holes) Soldier][10 350 cm (Holes) Soldier][10 400 cm (Holes) Soldier][10 450 cm (Holes) Soldier][10 500 cm (Holes) Soldier][10 550 cm (Holes) Soldier][10 600 cm (Holes)	6221177020613 6221177020620 6221177020637 6221177020644 6221177020651 6221177020668 6221177020675 6221177020682 6221177020699 6221177020705 6221177020729 6221177020736 6221177020743 6221177020750 6221177020767 6221177020774	15.60 21.20 26.50 31.80 37.10 42.40 47.40 53.00 58.30 63.60 74.20 84.80 95.40 106.00 116.60 127.20
	Splice 60 double	6221177022242	6.02
	Corner Splice	6221177022136	12.10

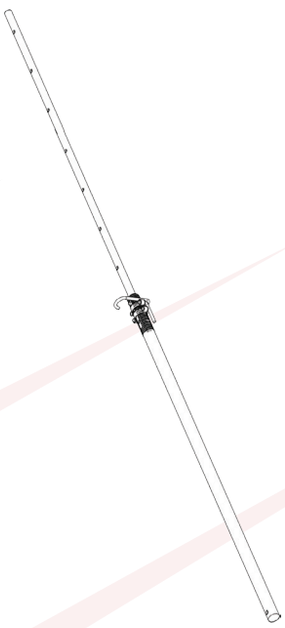
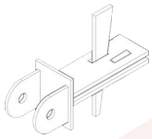
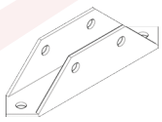
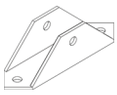
Components

Item	Description	Code no.	Weight (kg)
	Soldier][12 1.17 m (Slotted & Holes) Soldier][12 1.47 m (Slotted & Holes) Soldier][12 1.78 m (Slotted & Holes) Soldier][12 2.39 m (Slotted & Holes) Soldier][12 3.00 m (Slotted & Holes) Soldier][12 3.61 m (Slotted & Holes) Soldier][12 4.22 m (Slotted & Holes) Soldier][12 4.83 m (Slotted & Holes) Soldier][12 5.44 m (Slotted & Holes) Soldier][12 6.05 m (Slotted & Holes)	6221177020941 6221177020958 6221177020965 6221177020972 6221177033026 6221177020989 6221177020996 6221177021009 6221177021016 6221177021023	31.60 39.70 48.10 64.50 81.00 97.50 113.95 130.40 146.35 163.35
	Splice 99 cm	6221177022273	9.95
	Splice 147 cm	6221177032562	14.78
	Splice 211 cm	6221177032579	21.20
	Corner Splice (95 X 43) cm	6221177032692	13.86
	Corner splice (76 X 76) cm	6221177032708	15.30
	Pivot splice (60X60) cm Pivot splice (76X76) cm	6221177032715 6221177032722	12.10 16.10

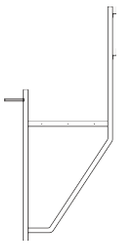
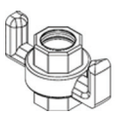
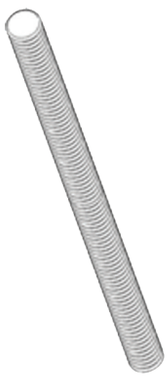
Components

Item	Description	Code no.	Weight (kg)
	Column corner splice	6221177032609	10.00
	Stop end connection	6221177032593	2.10
	Compression plate	6221177028039	1.28
	Universal corner angle tie	6221177026080	2.60
	Big wedge	6221177027629	0.85
	Small wedge	6221177027636	0.35
	Girder splice 60cm (single)	6221177022174	2.30

Components

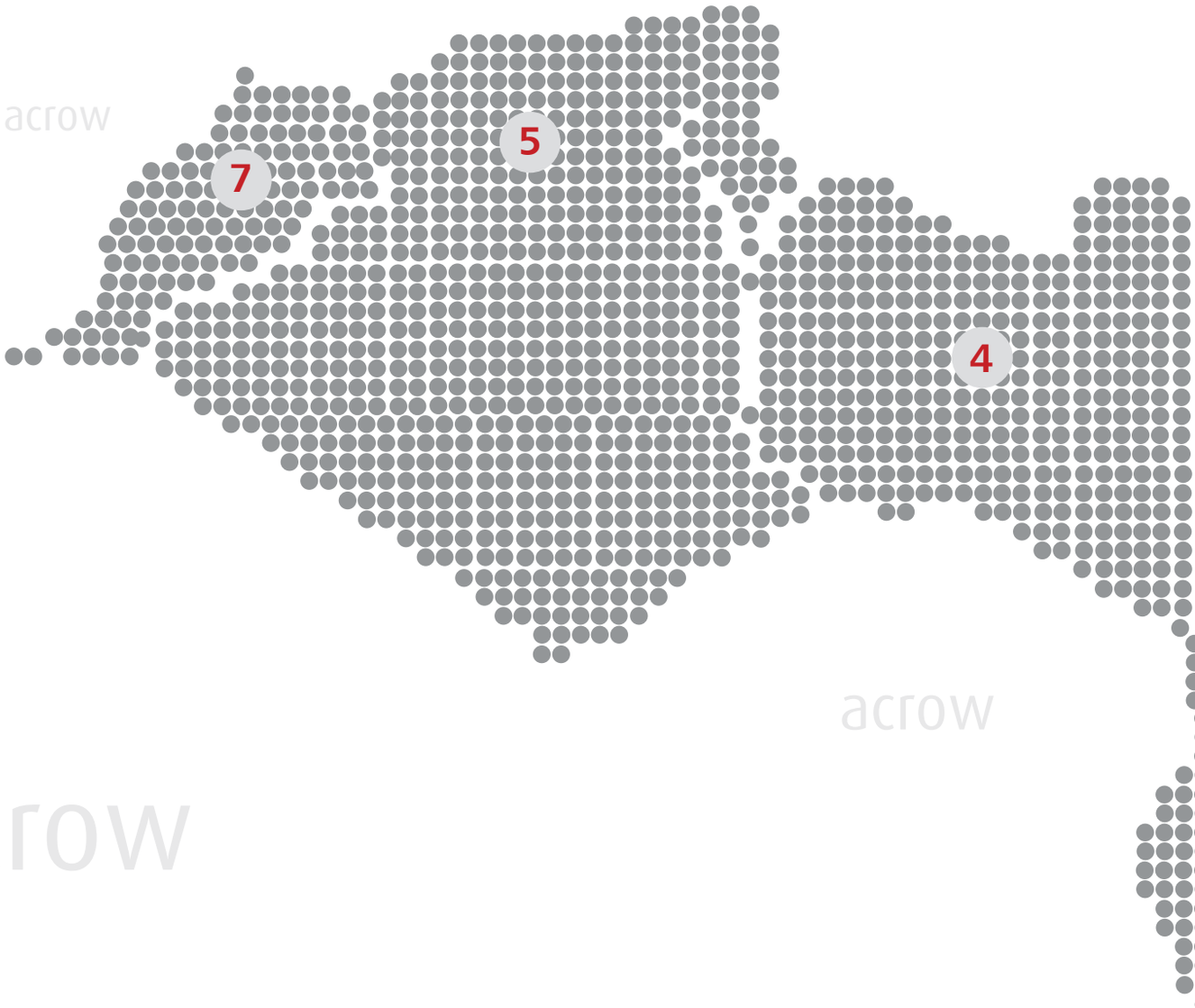
Item	Description	Code no.	Weight (kg)
	Tilt up shor X0 1.04 - 1.83 m Tilt up shor X1 1.75 - 3.20 m Tilt up shor X2 1.98 - 3.35 m Tilt up shor X3 2.95 - 3.96 m Tilt up shor X4 3.20 - 4.88 m	6221177023942 6221177023959 6221177023966 6221177023976 6221177023980	10.90 17.50 18.65 21.25 25.64
	Shore Connector for horizontal soldier	6221177006358	1.64
	Double Base plate	6221177000813	4.63
	Single Base plate	6221177000882	1.50
	Rivet pin L =7.50 cm Rivet pin L =12.50 cm	6221177012281 6221177012304	0.12 0.20
	Spring clip large Spring clip small	6221177022617 6221177022624	0.03 0.01

Components

Item	Description	Code no.	Weight (kg)
	Crane splice (4holes)	6221177022167	4.00
	Bolt m 16 X 80	2012090028596	0.16
	Nut m 16	2012210029779	0.033
	Assembling bracket	6221177003203	8.34
	Access bracket for horizontal soldier	6221177077167	21.00
	Wing nut DVD	6221177011383	0.44
	Tie rod DVD L=0.75 mm L=1.00 mm L=1.25 mm L=1.50 mm L=1.75 mm L=2.00 mm L=2.50 mm L=3.00 mm L=3.50 mm L=4.00 mm L=5.00 mm L=6.00 mm	 6221177023270 6221177023294 6221177023300 6221177023317 6221177023324 6221177023331 6221177023355 6221177023362 6221177023379 6221177023386 6221177023409 6221177023423	 1.19 1.58 1.98 2.37 2.77 3.16 3.95 4.74 5.50 6.32 7.90 9.48

Components

Item	Description	Code no.	Weight (kg)
	Steel washer	6221177027513	1.10
	Hi skaf Anchor	6221177028084	2.85
	Corner filer panel L shape H = 1.0 m Corner filer panel L shape H = 2.0 m Corner filer panel L shape H = 3.0 m	6221177006624 6221177006631 6221177006648	12.78 16.54 37.03
	Corner filer panel Lader shape H = 1.0 m Corner filer panel Lader shape H = 2.0 m Corner filer panel Lader shape H = 3.0 m	6221177006594 6221177006600 6221177006617	9.30 18.00 26.35



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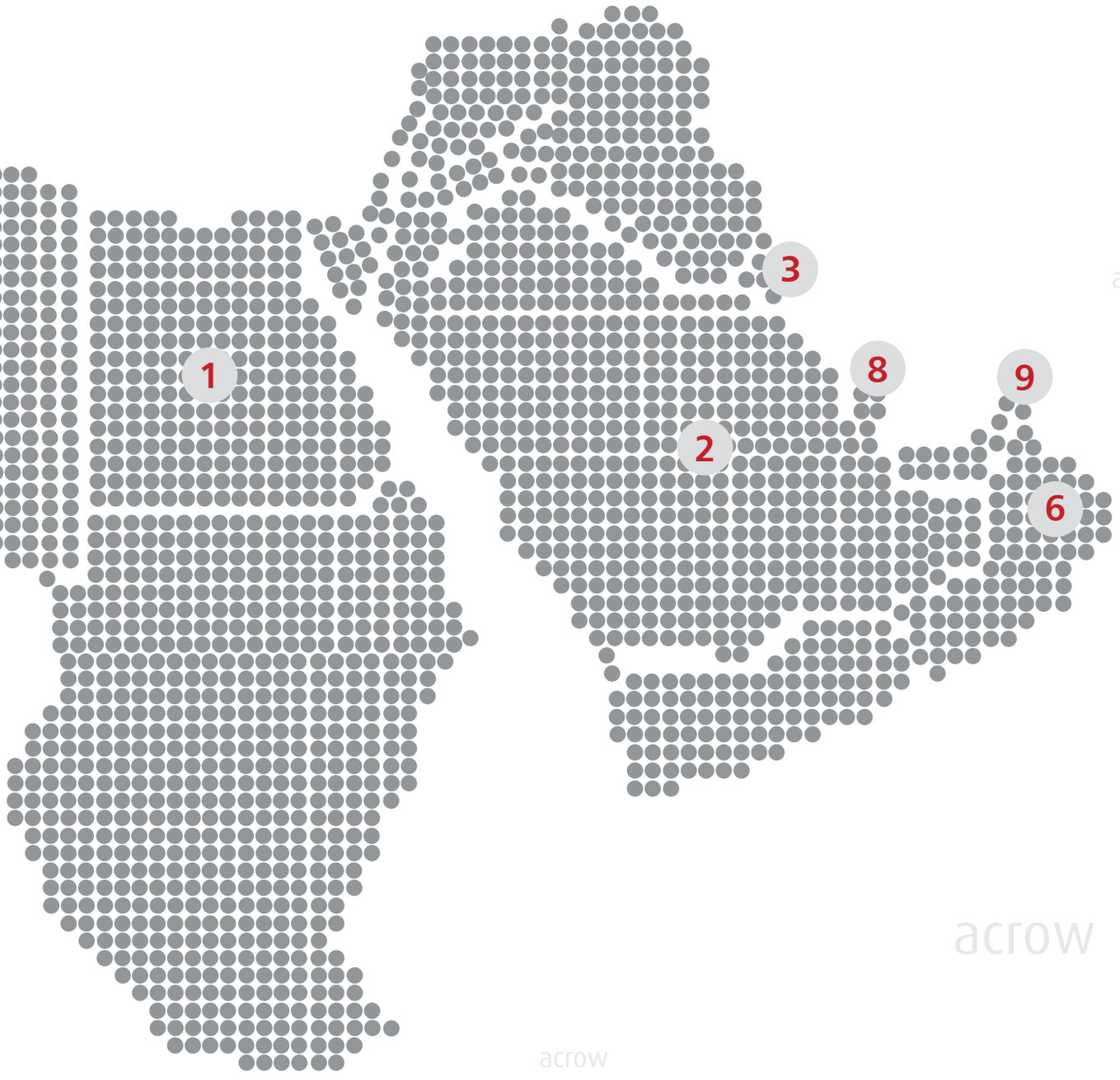

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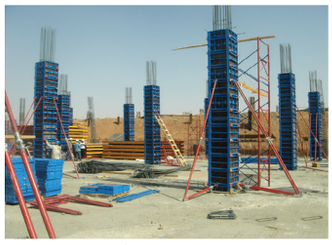
Formwork Technology

Acrow H20 & solider

Acrow One shutter solider


Acrow V.M.C

Acrow Tech-Form

Acrow Cuplock

Acrow Bridge table form

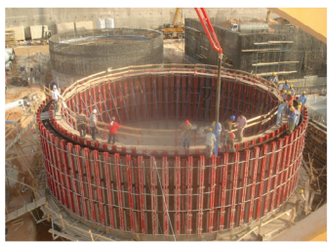
Acrow Circular column

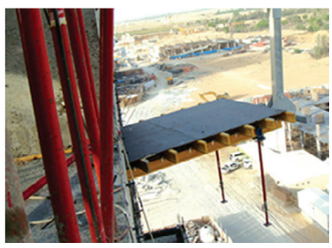
Acrow climbing system

Acrow shorbrace Table form

Acrow Big steel panel

Acrow European prop


Acrow Vari form


Acrow Prop table form

Acrow frame

Acrow early striking system

Acrow Ring lock

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